MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2025

SECOND SEMESTER

Discipline specific Core Course (DSC) – MG2DSCFNN100 – HOTEL HYGIENE AND FOOD SAFETY

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

Part A

Short Answer questions

Answer any **10** questions

Each question carries **2** marks

1. Which government body regulates food safety in India?

Ans: The **Food Safety and Standards Authority of India** (FSSAI) is the main government body responsible for regulating food safety in India. FSSAI sets standards for food products and oversees their manufacture, storage, distribution, sale, and import to ensure safe and wholesome food for consumers.

2. What is food borne illness?

Ans: Food borne illness are the diseases caused by agents entering human body through the ingestion of food. Most common symptoms are gastro-intestinal in nature. It is usually caused by bacteria, viruses, parasites, or chemical substances.

3. Why is food safety important for public health?

Ans: Food safety is important for public health because it helps prevent food-borne illnesses that can cause serious health problems or even death.

- Prevention of food borne illness.
- Protection of vulnerable individuals.
- Boosting of consumer confidence.
- Protection of producer.
- Improved Productivity.

4. What is the difference between cleaning and sanitization?

Ans: Cleaning- It is the process of removing visible dirt, dust, food residues from the surfaces.

Sanitization- It reduces the number of germs on surfaces to a safe level, involves using chemicals or heat after cleaning.

5. Define food safety and explain its importance?

Ans: Food safety is defined as an assurance that the food will not have any adverse effect on the consumer health when it is prepared or eaten according to its intended use.

Importance

- Prevention of food borne illness.
- Protection of vulnerable individuals.
- Boosting of consumer confidence.
- Protection of producer.
- Improved Productivity.
- Growth of food trade.

6. What is the risk associated with food hazards?

Ans: The risks associated with food hazards depend on the type of hazard, but they generally lead to serious health, economic and legal consequences.

- Health risk- can cause foodborne illnesses, chronic diseases and allergic reactions.
- Economic risk- it can affect the medical cost.

7. What is a chemical food hazard? Provide an example

Ans: Chemical hazard is any chemical contaminants introduced in food systems which may cause illness or injury to the individuals using the product.

Eg: Chemical pesticide residues on fruits and vegetables.

8. What is water activity (aw)?

Ans: Water activity is how much free water is available in a food product for microbial growth, chemical reactions and enzymatic activity. It is the ratio of the vapour pressure of water present in food substrate to the vapour pressure of pure water at the same temperature.

9. Define microorganism?

Ans: Microorganisms are present in food partly due to the natural microflora of raw material and additionally due to contamination during harvesting, processing, storage and distribution. They include a wide variety of life forms such as, Bacteria, Viruses, Fungi and Protozoa.

10. How moisture content affects the growth of microorganism?

Ans: Microorganism has an absolute demand for water, the exact amount of water needed for growth of microorganism varies. The water requirement of microorganism is expressed in physical form called Water activity.

11. Name two common bacterial pathogens that cause foodborne illnesses

Ans: Escherichia coli (E. coli) and Clostridium botulinum

12. Name two common adulterants found in milk.

Ans: Water and Detergent

13. What is a toxin-mediated foodborne illness?

Ans: It occurs when a person consumes food that contains harmful toxins produced by bacteria growing in the food. In this case, the illness is caused by the toxin, not directly by the bacteria themselves.

Eg: Clostridium perfingens, Bacillus cereus

14. How intoxication can be caused?

Ans: Food intoxication is caused when you eat food that already contains harmful toxins. These toxins are usually produced by certain bacteria, Molds or chemicals before the food is consumed.

15. Write any 3-detection method for common adulterants in food.

Ans:

- Detection of water in milk – Add a few drops of milk into a slanting surface, flows immediately without leaving a white trail it shows the presence of water
- Detection of starch in milk- Add a few drops of iodine solution to a milk sample, If the milk turns blue, it indicates the presence of starch.
- Detection of sugar in honey- Add a few drops of honey to the glass containing water, it disperses in water shows the presence of added sugar.

Part B

Short Essay questions

Answer any **4** questions

Each question carries **5** marks

16. What are the key factors determine the quality of food? Explain any five.

Ans: The quality of food is determined by various factors that ensure it is safe, nutritious, and acceptable to consumers. Here are five key factors:

- Nutritional Value- Presence of essential nutrients such as carbohydrates, proteins, fats, vitamins, and minerals in the right proportions.
- Freshness- Fresh food is free from spoilage and degradation. It has a natural flavor, aroma, and texture.
- Hygiene and Safety- Food should be free from harmful bacteria, viruses, chemicals, and physical contaminants.

- Taste and Appearance- Sensory qualities such as color, flavor, texture, and aroma play a major role in consumer acceptance.

17. Discuss the health hazards caused by chemical contaminants in food?

Ans: Chemical contaminants in food are harmful substances that can enter the food supply during farming, processing, packaging, or storage. These contaminants pose serious health hazards.

- Acute Poisoning- High levels of chemicals such as pesticides or cleaning agents can cause immediate symptoms like nausea, vomiting, dizziness, abdominal pain, and even death in extreme cases.
- Chronic effect- long term exposure to low level of chemicals can lead to serious health problems like cancer, kidney damage and neurological disorders.
- Organ damage- Exposure to certain chemicals can damage specific organs like liver or kidney.
- Developmental effects- In children, exposure to chemicals can impacts mental intellectual and physical level.
- Reproductive issues- Chemical exposure may lead to reproductive problems and birth defects.

18. How food infection can be defined?

Ans: Food infection refers to a type of foodborne illness that occurs when a person consumes food or water contaminated with pathogenic microorganisms, such as bacteria, viruses, or parasites.

- Cause- The ingestion of live microorganisms such as *Salmonella*, *E. coli*.
- Mechanism- After consumption, the pathogens colonize the gastrointestinal tract and start reproducing, leading to the release of toxins or causing direct damage to the tissues.
- Symptoms- Fever, diarrhoea, abdominal cramps, nausea, and vomiting.
- Sources- Contaminated water, undercooked meat, unpasteurized milk, raw vegetables, and unhygienically prepared food.
- Prevention- Proper cooking, personal hygiene, safe food handling, and clean water supply.

19. Explain morphological characteristics of bacteria?

Ans: Bacteria are microscopic, single-celled organisms that vary in shape, size, and arrangement.

- Shape- Cocci (Spherical), Bacilli (Rod), Spirilla (Spiral or helical), Vibrios (Comma)
- Size- Range from 0.2 to 2.0 micrometers in diameter and 1 to 10 micrometers in length. Cocci are generally smaller than bacilli.
- Flagella and Motility- Bacteria have flagella, which are tail-like structures used for movement.
- Cell Wall Structure.

20. Describe incidental adulteration and its sources in food processing.

Ans: Incidental adulteration refers to the unintentional contamination of food products that occurs during various stages of food production, processing, packaging, transportation, or storage.

- Contaminated Equipment and Machinery- Adulterated due to rust, grease, or metal particles from poorly maintained machines or processing equipment.
- Improper Storage Conditions- Food stored in damp, unhygienic, or infested warehouses.
- Use of Contaminated Water- Using unclean or chemically contaminated water during washing, processing, or mixing.
- Pesticide and Chemical Residues- Excessive or improper use of pesticides during farming can leave chemical residues.
- Packaging Materials- Food may get contaminated through chemicals from low-grade plastic or recycled packaging materials, which can leach into the food during storage.

21. Explain how food adulteration affects the economy and public health.

Ans: Food adulteration has serious consequences for both public health and the economy. It involves the addition of inferior or harmful substances to food, which can lead to widespread harm and loss.

- Health Hazards and Diseases- cause a range of health issues such as food poisoning, cancer, liver and kidney damage, neurological disorders, and in severe cases, death.
- Malnutrition- Reduce the nutritional quality of food, leading to deficiencies and weakened immunity, especially in vulnerable groups like children and the elderly.
- Long-Term Chronic Illnesses- Continuous consumption of contaminated food can result in chronic illnesses such as hormonal imbalance, reproductive problems, and developmental disorders.
- Loss of Consumer Trust- Damages consumer confidence in domestic and export food markets.

Part C

Essay questions

Answer any **1** question

Each questions carries **10** marks

22. Define food hygiene and explain its importance in ensuring food safety.

Ans: Food Hygiene refers to the conditions and practices necessary to ensure the safety, cleanliness, and wholesomeness of food at all stages—from production and processing to preparation and consumption. This includes proper handling, preparation, storage, and distribution of food to prevent contamination and foodborne illnesses.

- Prevention of Foodborne Diseases- Good food hygiene practices help eliminate harmful bacteria, viruses, and parasites that can cause serious health problems like food poisoning, diarrhoea, and infections.
- Promotion of public health- Proper food hygiene practices reduce spread of infectious disease, especially in communities and food establishments.
- Enhanced Food quality and safety – Sanitation ensures that food is free from contaminants maintaining its nutritional value, flavor and quality.
- Compliance with regulations- Adhering to food safety laws and standards, avoid legal consequences, penalties and laws of reputations, consumer trust.
- Economic benefits – Reduces healthcare host and loss productivity.
- Protection of vulnerable – Children, pregnant women elderly and those with the weakened immune systems are particularly risk from food borne illness, making hygienic conditions is essential for those populations.
- Environmental impact- Proper waste management in food production and preparation minimizes environmental pollution, contributing to sustainability.

23. Write a detailed note on how relative humidity and water activity affects the growth of microorganism?

Ans: The growth and survival of microorganisms in food are heavily influenced by moisture-related factors, primarily relative humidity (RH) and water activity (aw).

Relative Humidity- Relative Humidity is the amount of moisture present in the air compared to the maximum it can hold at a given temperature, expressed as a percentage.

- Impact on Food- RH affects the moisture content of foods during storage. If the RH is too high, dry foods may absorb moisture, tends to microbial growth.
- Microbial Growth- Molds and some bacteria can grow on food surfaces when RH is high.
- Control Measures- Proper packaging and environmental control help maintain optimal RH and minimize microbial risks.

Water Activity- Water activity is a measure of the free or available water in a food product for microbial use.

- Different microorganisms have minimum aw requirements for growth.
- Lowering aw through drying, freezing, or adding solutes like salt and sugar is a common method to preserve food and prevent microbial spoilage

24. What are the preventive measures to control food borne pathogens?

Ans: Foodborne pathogens such as Salmonella, E. coli, Listeria, and Norovirus are responsible for millions of food-related illnesses globally. Preventing contamination and controlling these pathogens is essential for ensuring food safety.

- Personal Hygiene- Regular handwashing with soap and clean water, especially after using the toilet or handling raw food. Use of clean uniforms, gloves, and hairnets to prevent contamination.

- Proper Cooking and Temperature Control- Cook food to the correct internal temperature to kill pathogens.
- Safe Storage and Refrigeration- Store perishable items in the refrigerator or freezer. Avoid leaving cooked or perishable food at room temperature for more than 2 hours.
- Cross-Contamination Prevention- Use separate cutting boards, utensils, and containers for raw and cooked foods.
- Cleaning and Sanitization- Regularly clean and sanitize kitchen surfaces, equipment, and utensils.
- Waste management- Proper disposal of food waste to avoid attracting pests or causing contamination.
- Pest Control- Keep food storage and preparation areas clean and sealed. Implement pest control programs.